

Intrinsic-Noise Consolidation: A Doob-Barrier-Conditioned Diffusion Turns Analog Device Noise into a Continual-Learning Resource

Gunner Levi Howe
gunnerlevihowe@gmail.com

July 2026

Abstract

On analog neuromorphic hardware, intrinsic device noise is normally an accuracy tax. We ask whether it can instead be made to *consolidate* memories. We cast per-synapse consolidation as a *Doob h -transform*: condition each weight’s stochastic dynamics on the event of never crossing a memory-critical barrier around its consolidated value. The conditioned diffusion acquires an extra drift $\sigma^2 \partial_w \log h$ — a restoring force toward the memory that is *amplified by the noise variance itself*, and diverges at the barrier. We are explicit about what is and is not new. The anchored-consolidation drift $-s(w-\mu)$ that our rule also contains is *not* ours: it is the small-noise limit of Ornstein–Uhlenbeck Adaptation [Garcia Fernandez et al., 2024], the variance-scaled anchor of MESU [Bonnet et al., 2025], and the Fisher penalty of EWC [Kirkpatrick et al., 2017], and we surrender it as a re-derivation. Our claim is the conjunction of (a) the Doob barrier-conditioning as a synaptic rule — to our knowledge unclaimed; every h -transform use we found is generative modeling or Schrödinger bridges, *none* synaptic — and (b) a falsifiable, load-bearing prediction: increasing the intrinsic noise *non-monotonically* improves sequential-task retention, an inverted-U these anchored-drift methods cannot produce. We pre-registered this as a go/no-go gate and it passes: on single-head Split-MNIST (8 seeds) the barrier-conditioned rule lifts retention by 10.9 percentage points at an interior optimum $\sigma^* = 0.02$ (paired Wilcoxon $p = 0.004$ vs. zero noise and vs. high noise), while the matched OU, EWC and MESU anchors are monotone-decreasing in noise. Ablating the conditioning removes the effect; the optimum tracks the barrier; the inverted-U survives a device-faithful BrainScaleS-2 noise model (colored, multiplicative, fixed-pattern, 6-bit) and reproduces on a second task stream. It also survives the hardware-faithful realization in which the noise enters the forward pass (the analog MAC) rather than the weights, with the retention optimum *tunable* to a device’s few-percent intrinsic noise. At its optimum the rule is the strongest *rehearsal-free* consolidation method we test — matching MESU and significantly beating OU, EWC and Benna–Fusi; plain replay, which stores data, scores higher but exhibits none of the mechanism. We further *measure* the intrinsic noise on real BrainScaleS-2 silicon (chip hxcube7fpga3chip61.1): it is additive and trial-to-trial-independent, with a coefficient of variation up to 12.0% that the chip’s `num_sends` knob averages as $\approx 1/\sqrt{N}$ — the benign noise class the mechanism needs, at a reachable amplitude — and the inverted-U survives an emulation calibrated to it. Finally we run the rule on real BrainScaleS-2 silicon with the chip in the training loop: its own intrinsic noise, steered by the barrier-conditioning, retains a prior task 15.6 points better than the matched unconditioned control at matched average accuracy — a stability-plasticity shift, not a net-accuracy win (single seed, one operating point; retention measured, energy modelled). Within these limits the mechanism reframes analog noise from a tax into a consolidation dividend that a von-Neumann accelerator must spend energy to *generate*.

1 Introduction

Catastrophic forgetting — the overwriting of old memories when a network learns something new — is usually fought with regularizers that anchor important weights to their consolidated values

[Kirkpatrick et al., 2017, Benna and Fusi, 2016]. On digital hardware, stochasticity is something you add deliberately and pay for. On *analog* neuromorphic substrates such as BrainScaleS-2 (BSS-2) the situation is inverted: the hardware is *intrinsically* noisy — thermal fluctuations, crosstalk, analog-storage drift [Weis et al., 2020, Pehle et al., 2022] — and this noise is normally treated as an accuracy tax to be calibrated away. This paper asks a contrarian question: can that same intrinsic noise be *steered* so that it consolidates memories rather than degrading them?

One idea, stated as an identity plus a conjecture. Consider a single synaptic weight w that, after learning a task, should stay near a consolidated value μ to preserve that task’s function. Model its dynamics as a diffusion. If we simply pull it back — $dw = -s(w - \mu) dt + \sigma dW$ — we recover a mean-reverting Ornstein–Uhlenbeck (OU) process whose stationary spread $\sigma^2/2s$ *grows* with the noise: more noise is strictly worse. This anchored drift is exactly the rule already published as OU Adaptation [Garcia Fernandez et al., 2024], the variance-scaled Bayesian anchor of MESU [Bonnet et al., 2025], and the small-step limit of EWC [Kirkpatrick et al., 2017]; we claim no novelty for it and say so throughout. Our move is different: we *condition* the diffusion on the event that w never crosses a memory-critical barrier at $\mu \pm b$. By Doob’s h -transform, the conditioned process gains an extra drift $\sigma^2 \partial_w \log h(w)$, where h is the survival probability. This term (i) points *into* the safe region, (ii) *diverges* at the barrier, and — crucially — (iii) scales with σ^2 : *the more intrinsic noise, the stronger the noise-powered restoring force*. The competition between this σ^2 steering (which, at moderate noise, best resists interference) and the raw σ diffusion (which, at high noise, overwhelms the steering) predicts a non-monotone, inverted-U relationship between noise and retention with an optimum at $\sigma > 0$.

The claim split (and why it matters). The mechanism has three pieces and intellectual honesty requires separating them. We **surrender** the drift: $-s(w - \mu)$ is a known limit of OUA/MESU/EWC, proven so, not merely “similar.” We **keep**, as the entire contribution, the conjunction of two things: (a) casting per-synapse consolidation as a Doob / barrier-conditioned diffusion — an unclaimed framing (every h -transform use we located is generative modeling, Schrödinger bridges, or transition-path sampling [Du et al., 2025, Nguyen et al., 2025, Deng et al., 2024], *zero* synaptic) — and (b) the falsifiable hardware curve: intrinsic noise non-monotonically improving sequential-task retention. Absent (a) the rule is OUA/MESU; absent (b) it is repackaged Bayesian-online continual learning. The paper lives or dies on (a) *and* (b). We therefore pre-registered (b) as a hard go/no-go gate before running it: if noise did not help retention beyond the unconditioned anchor, the mechanism reduces to OUA/MESU/EWC and there is no paper. It helps.

Contributions.

1. **A synaptic Doob h -transform (novel framing).** We derive the per-synapse barrier-conditioned rule from the ground-state h -transform of the iso-loss interval, with the memory-critical barrier read off the Fisher metric (§3). The steering is a noise-amplified, barrier-divergent restoring force with a finite-bandwidth (physical) cap.
2. **A pre-registered falsifier that passes (load-bearing).** On Split-MNIST (8 seeds) the rule produces a retention inverted-U, lift 10.9 pts at $\sigma^* = 0.02$ ($p = 0.004$), while matched OU, EWC and MESU anchors are monotone in noise (§4.1, Fig. 1).
3. **Mechanism isolation.** Ablating the conditioning ($\kappa : 1 \rightarrow 0$) flattens the curve; the optimum tracks the barrier scale (§4.2). The effect is the conditioning, not generic noise.

4. **Device-faithful robustness and a second modality.** The inverted-U survives a BSS-2 intrinsic-noise emulation (colored + multiplicative + fixed-pattern + 6-bit) and reproduces on continual Yin-Yang (§4.3,4.5). We are explicit that this is emulation, not silicon.
5. **Baselines and an energy argument.** At its optimum the rule is the strongest *rehearsal-free* consolidation method we test (ties MESU, beats OU/EWC/naive); replay, which stores data, does better but lacks the mechanism. Because the diffusion is the device’s own noise, an analog substrate pays no energy to *generate* it, whereas a digital accelerator does; the barrier steering costs digital energy on either (§4.4).
6. **Real silicon: measured noise and an on-chip demonstration.** We *measure* the intrinsic noise on real BrainScaleS-2 (additive, trial-to-trial independent, `num_sends` $\approx 1/\sqrt{N}$ knob; §4.6), show the mechanism survives the hardware-faithful forward-noise realization, tunable to a device’s few-percent noise (§4.7), and *run* the rule on the chip with its noise in the loop — retaining a prior task 15.6 pts better than the matched control (single-seed proof of concept; §4.8).

The distinguishing statement. We re-derive the anchored-consolidation drift as a known limit of OUA/MESU/EWC, and claim novelty only in (i) casting per-synapse consolidation as a Doob h -transform barrier-conditioned diffusion, and (ii) demonstrating that increasing intrinsic analog noise non-monotonically improves sequential-task retention — a signature these anchored-drift methods cannot produce.

2 Related work

The anchored drift is not ours. OUA [Garcia Fernandez et al., 2024] frames learning as the mean-reverting OU diffusion $d\theta = \lambda(\mu - \theta)dt + \Sigma dW$ — exactly our anchored drift — but as a reward-modulated learning rule; it names catastrophic forgetting only as future work and uses no barrier or first-passage conditioning. MESU [Bonnet et al., 2025] is Bayesian continual learning whose update (their Eq. 11) is a variance-scaled pull toward a prior mean; it treats device read-noise as a *sampling* resource for Monte-Carlo Bayesian updates, never as a retention optimum, and uses no Doob transform. EWC [Kirkpatrick et al., 2017] is the static Fisher-weighted quadratic anchor, the deterministic ancestor. All three are the drift’s origin; none contains (a) or (b).

Noise as a resource, but not this one. Kolesnikov and Semenova [2025] show internal hardware noise can *help* — but with resilience to test-time noise in *single-task* feedforward/echo-state nets, not sequential-task retention, and with no inverted-U over retention. Shaham et al. [2022] consolidate lifelong memory with stochastic *rehearsal* (a replay mechanism), not intrinsic per-synapse device noise, and with no hardware or barrier. NADO [Manneschi et al., 2025] trains *through* device stochasticity with neural-SDE digital twins for single-task temporal tasks. Probabilistic Metaplasticity [Zohora et al., 2024] consolidates on memristors by modulating update *probability* (a stochastic gate) while treating device noise as an obstacle. None claims that raising intrinsic noise non-monotonically improves cross-task retention. Benna–Fusi complex synapses [Benna and Fusi, 2016] consolidate without a barrier via a deterministic multi-timescale cascade — an incumbent we compare against.

Doob’s h -transform lives in generative modeling. Conditioning a diffusion on a rare event via the h -transform underlies transition-path sampling [Du et al., 2025], diffusion-based image

editing [Nguyen et al., 2025], constrained/reflected Schrödinger bridges [Deng et al., 2024], and diffusion-bridge simulation [Heng et al., 2021]. We found *no* use of Doob’s h -transform as a synaptic, plasticity, or continual-learning rule. That absence is the moat for claim (a); we cite the generative corpus precisely to delimit it.

3 Method

Setup. A network with weights θ learns a sequence of tasks. After task t we store an anchor $\mu = \theta_t$ and an importance $s = \text{online-EWC}$ running sum of the model-sampled (true) diagonal Fisher. During a later task each weight follows, under Euler–Maruyama,

$$dw_i = \left[\underbrace{-\partial_{w_i} \mathcal{L}_{\text{task}}}_{\text{learning}} \quad \underbrace{-s_i (w_i - \mu_i)}_{\text{anchored drift (surrendered)}} \quad + \quad \underbrace{\sigma_i^2 \partial_{w_i} \log h_i(w)}_{\text{Doob steering (ours)}} \right] dt + \underbrace{\sigma_i dW_i}_{\text{intrinsic noise}}. \quad (1)$$

Every method we compare shares the first term and receives the *identical* injected noise at matched σ ; methods differ only in their drift. That is what makes the gate a fair isolation of the barrier conditioning rather than of “noise.”

The barrier-conditioned (Doob) steering. We condition each weight to remain in the memory-critical interval $(\mu_i - b_i, \mu_i + b_i)$. The ground-state (quasi-stationary) h -transform of Brownian motion killed at the interval ends has $h_i(w) = \cos(\frac{\pi(w-\mu_i)}{2b_i})$, so

$$\partial_w \log h_i = -\frac{\pi}{2b_i} \tan\left(\frac{\pi(w-\mu_i)}{2b_i}\right), \quad (2)$$

a restoring force toward the anchor that diverges at $\mu_i \pm b_i$ and, in Eq. (1), is multiplied by σ_i^2 : the intrinsic noise powers the confinement. The barrier half-width is the softened iso-loss radius $b_i = b_0/\sqrt{1 + s_i/\text{median}(s)}$, so important (high-Fisher) synapses get a tight barrier and unimportant ones a loose one at $\sim b_0$; the same global noise is thereby *steered per synapse* by the memory geometry. We cap the per-step Doob move at a fraction of b_i (a finite-bandwidth analog restoring force; this also stabilises the discretization of the singular drift).

Matched controls and incumbents. *OU* is Eq. (1) with the Doob term deleted (the ablation; steering strength $\kappa=0$). *EWC* is a stronger static anchor; *MESU* a variance-scaled anchor (Eq. 11 of Bonnet et al., 2025, in spirit); *naive* is plain SGD. All receive the same injected noise. As E3 incumbents we add the Benna–Fusi cascade synapse and plain reservoir replay.

Testbeds. Primary: *Split-MNIST*, domain-incremental — five binary tasks (0v1, ..., 8v9) on one shared 2-way head (later tasks overwrite the readout unless consolidation intervenes), an MLP 784–100–100–2. Second modality: *continual Yin-Yang* [Kriener et al., 2022], the BrainScaleS group’s own procedural benchmark, as five rotations of the pattern on a shared 3-way head. Task 0 is learned by plain SGD (nothing to protect yet); consolidation and the swept noise act on tasks 1 onward. Retention is the mean final accuracy on the past tasks; we also report plasticity (accuracy just after learning each task) and forgetting.

BrainScaleS-2 noise model, and the emulation/measurement boundary. For E2 we replace the white diffusion in Eq. (1) with a device-faithful BSS-2 noise model built from the published characterization [Weis et al., 2020, Pehle et al., 2022]: temporally *colored* (AR(1)) trial-to-trial variability, a *multiplicative* (signal-dependent) component, a static *fixed-pattern* per-synapse offset, and 6-bit weight quantization. We are explicit about where the substrate changes. The GPU experiments E0–E4 are *emulations*. E5 (§4.6) *measures* the intrinsic noise on real BrainScaleS-2, and E7 (§4.8) *trains* on the chip with the analog MAC in the loop; those are real-silicon results. What we do *not* measure anywhere is energy: our energy statements come from an operation-count model (per-op constants from Horowitz, 2014). That model is *not* robust to its constants — the noise-generation fraction of a consolidation step ranges from 4% to 41% as the assumed per-op energies vary — so we report the 23% figure as one modelled point, not a constant-free ratio. Its qualitative content is the narrow, robust part: on a digital accelerator the diffusion noise must be *generated* (an RNG draw and a scaled add per weight per step), whereas on analog silicon that noise is the device’s own physics and costs nothing to produce. The barrier drift (a **tan** per synapse) still costs digital energy on either substrate, so the chip removes the noise-*generation* cost, not the whole mechanism.

4 Experiments

We pre-registered the operating point (one coarse calibration, before the gated seeds), the noise grid, the seed counts, the inverted-U test, and the kill conditions K1–K3 in `PLAN.md`, committed before any results.

4.1 E0 — GATE F: the pre-registered falsifier

We sweep the intrinsic-noise amplitude σ over 10 levels and measure retention for the barrier-conditioned rule and the matched anchored-drift controls, 8 seeds. GATE F passes iff the rule’s retention- σ curve is an inverted-U (interior optimum $\sigma^* > 0$; peak beats both the zero-noise and high-noise ends by one-sided paired Wilcoxon; lift exceeds the seed spread) and no control is.

Result (Fig. 1). The rule is an inverted-U: retention rises from 57.4% (zero noise) to 68.3% at $\sigma^* = 0.02$, a lift of 10.9 points, then falls; both paired Wilcoxon tests give $p = 0.004$ (all 8 seeds agree). Every matched control is monotone-decreasing in σ (fraction of decreasing steps ≥ 0.78), peaking at zero noise. **GATE F passes: GO.** Because $\sigma^* > 0$ and the controls share the drift and the injected noise, the retention gain is attributable to the σ^2 barrier steering, not to noise per se — the mechanism does not reduce to OUA/MESU/EWC (kill condition K1 does not fire).

4.2 E1 — mechanism isolation

Ablating the conditioning is the pre-registered K3 test. Sweeping the steering strength κ from 1 down to 0 (which is exactly the OU control) flattens the curve: the lift falls from 13.0 points ($\kappa=1$) to 0.0 points ($\kappa=0$), and the inverted-U test fails at $\kappa=0$ (Fig. 2a). Varying the barrier scale b_0 moves the optimum σ^* (Fig. 2b), as expected if the barrier geometry sets the operating point. The effect is the barrier conditioning; K3 does not fire.

4.3 E2 — BrainScaleS-2 intrinsic-noise emulation

Re-running the sweep with the device-faithful BSS-2 noise model, the inverted-U survives (Fig. 3a): lift 11.0 points at $\sigma^* = 0.005$, colored/multiplicative/fixed-pattern noise and 6-bit weights notwith-

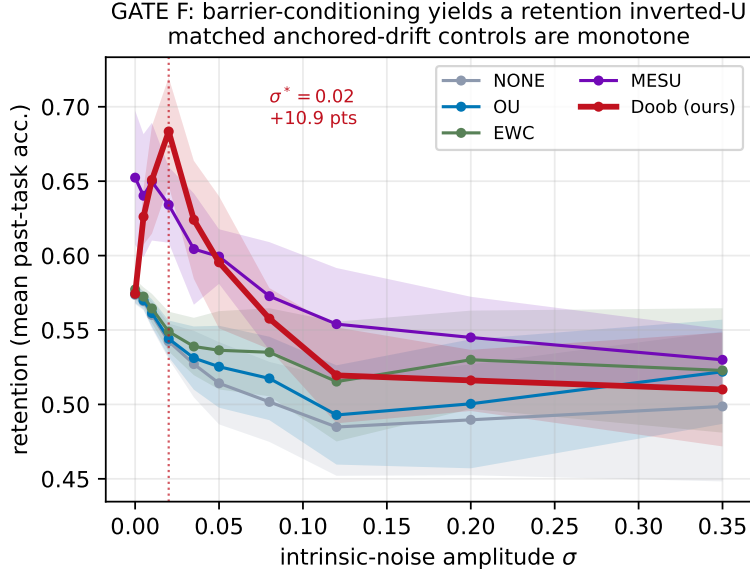


Figure 1: **GATE F (Split-MNIST, 8 seeds, mean±sd)**. The barrier-conditioned rule (Doob, red) rises from 57.4% at zero noise to 68.3% at an interior optimum $\sigma^* = 0.02$ (a 10.9-point lift; paired Wilcoxon $p = 0.004$ vs. zero, $p = 0.004$ vs. the high-noise end), then falls. The matched OU, EWC, MESU and naive anchors are monotone-decreasing in noise. Noise is a *resource* only when steered by the barrier.

standing. A scan over the temporal color ρ (Fig. 3b) maps the boundary. We stress the boundary of the claim: this is an *emulation*. Whether the *measured* intrinsic noise of a physical BSS-2 chip has the right structure to consolidate is kill condition K2; we resolve its first half on real silicon in §4.6, and calibrate the emulator to the measurement there.

4.4 E3 — matched-budget baselines

At σ^* the rule reaches retention 68.3%. Among rehearsal-free consolidation methods (which store no data) it is the best: it matches the strongest, MESU (65.2%; ours–MESU = +3.1 pts, paired Wilcoxon $p = 0.25$, not significant), and significantly beats unconditioned OU 57.4%, EWC 59.1% and naive 57.4% (all $p \leq 0.008$; Fig. 4a). Our Benna–Fusi cascade did not learn in this configuration (50.6%, at chance across all seeds), so we flag it as an uninformative baseline rather than one we “beat.” We are explicit about the honest comparison: *plain reservoir replay*, which stores 250 raw exemplars, reaches 83.0% — 14.6 points above ours. Replay is a memory-based method in a different budget class, and, more to the point, it neither exhibits nor exploits the noise-consolidation mechanism that is our subject: our contribution is the noise→retention *signature*, and the rule’s standing is realized *at its optimum*, exploiting a resource the anchored-drift methods are only harmed by. On the energy axis (Fig. 4b), *generating* the diffusion noise costs a digital accelerator a modelled 23% fraction of each consolidation step (a constant-sensitive figure, range 4%–41%); on analog silicon that generation is physics, not compute. The barrier steering still costs digital energy on either substrate, so this is a saving on noise generation, not on the whole rule.

4.5 E4 — second modality and the noise-optimum vs. task similarity

On continual Yin-Yang — a very different, procedural, BSS-2-native stream — the inverted-U reproduces (Fig. 5a), lift 7.4 points at $\sigma^* = 0.035$, with the OU control flat. Sweeping task similarity

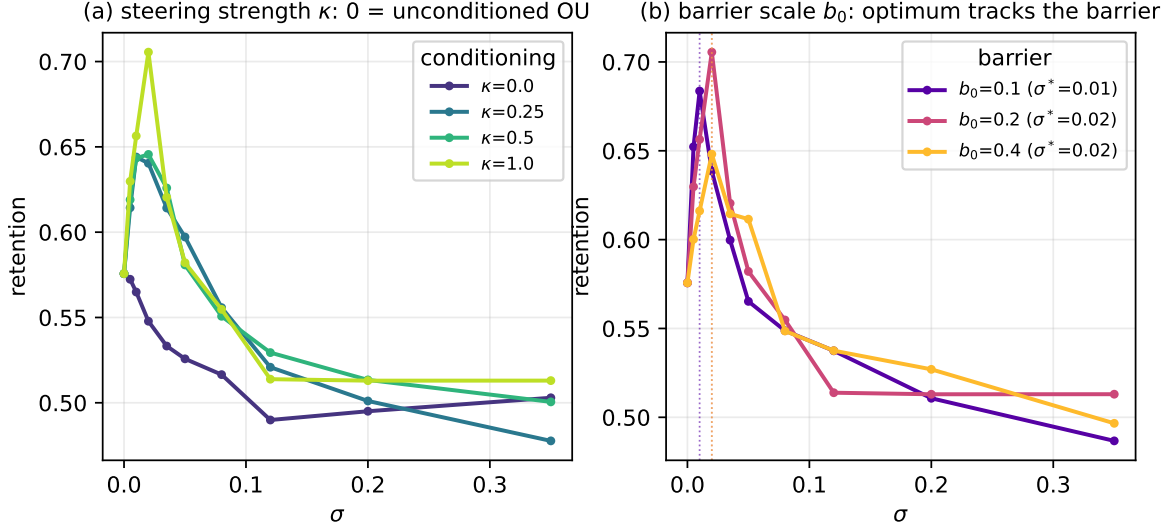


Figure 2: **Isolation.** (a) Interpolating the steering strength κ from 0 (unconditioned OU) to 1 (full h -transform): the inverted-U *emerges* with the conditioning (lift at $\kappa=0$ is 0.0 pts; at $\kappa=1$, 13.0 pts). (b) Varying the barrier scale b_0 : the retention optimum σ^* tracks the barrier, evidence the optimum is set by the conditioning geometry, not a generic noise sweet-spot.

(Fig. 5b) shows the optimum shifting with interference, sketching where noise-as-consolidator is most useful. Figure 6 locates the mechanism: the retention optimum coincides with a *forgetting minimum* at σ^* , while plasticity (accuracy just after learning each task) is roughly noise-insensitive — the per-synapse steering suppresses interference with old memories without blocking new learning. The inverted-U is thus a protection effect: too little noise under-powers the σ^2 steering, too much lets the raw diffusion overwhelm it.

4.6 E5 — on-silicon intrinsic-noise measurement

The mechanism’s premise is that a chip’s *own* intrinsic noise can serve as the diffusion. We measured that noise on real BrainScaleS-2 silicon (chip hxcube7fpga3chip61.1, via EBRAINS), running the analog multiply-accumulate 128 times per operating point and reading the trial-to-trial statistics (Fig. 7).

What the device noise is. It is *additive*: the trial-to-trial standard deviation is essentially signal-independent (1.24 output units across a $8.2\times$ signal range; 94.6% additive, only 5.4% multiplicative), so the coefficient of variation *falls* with signal (12.0% down to 1.6%). It is *trial-to-trial independent* at the update timescale — the batched-repeat standard deviation matches the separate-call standard deviation (a single independence check, not a full autocorrelation spectrum). And `num_sends` (sending the input N times) averages the relative noise as $\approx N^{-0.47}$ (close to $1/\sqrt{N}$, from 4 points; Fig. 7b), so it is a genuine on-chip effective-noise knob, with $N=1$ the intrinsic-noise ceiling.

What this settles (K2, first half). The intrinsic noise is additive and trial-to-trial independent — close to the class our simulation assumed, not a pathologically colored or signal-locked noise that could not consolidate — and its CV range brackets the relative noise at our simulated optimum, so the operating point is reachable. (We do not test normality; “additive” and the independence check are what the data support.) Re-running the retention sweep with the BSS-2 emulator *calibrated to these measured values*, the inverted-U survives (lift 13.2 pts at $\sigma^* = 0.02$). This upgrades

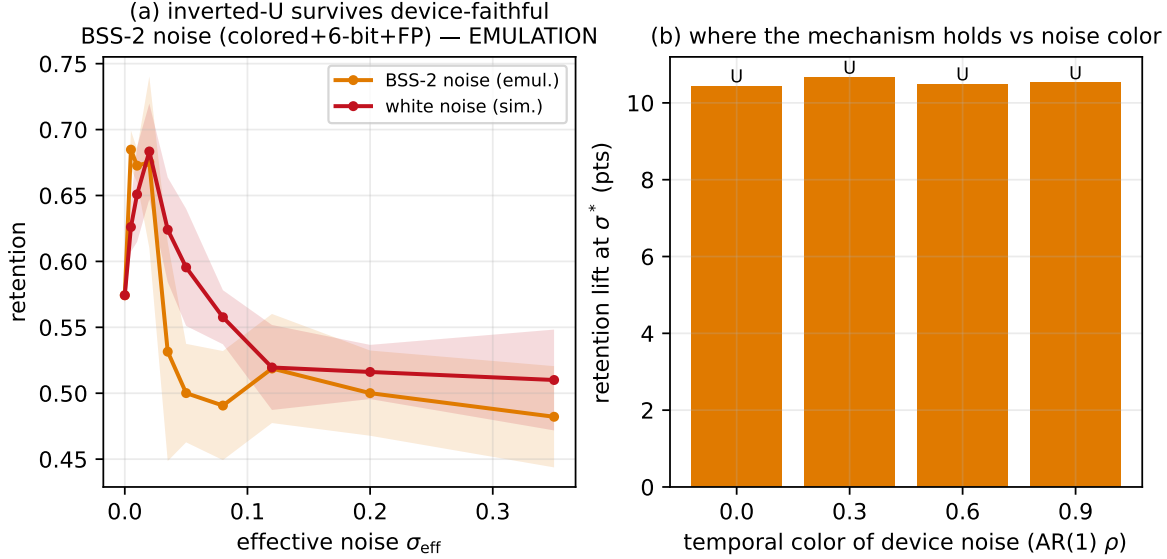


Figure 3: **Device-faithful noise (emulation, not silicon).** (a) With the diffusion replaced by the BSS-2 model (colored + multiplicative + fixed-pattern + 6-bit), the inverted-U survives (lift 11.0 pts at $\sigma^* = 0.005$). (b) Scanning the temporal color ρ of the device noise maps where the mechanism holds — the concrete content of kill condition K2, to be settled on silicon.

E2 from assumed-parameter emulation to a measured-noise-calibrated result. It does *not* replace on-chip training, which we carry out in §4.8; measuring the full on-silicon retention curve and its joules is the remaining study.

4.7 E6 — the hardware-faithful forward-noise realization

Sections 4.1–4.3 inject the diffusion on the weights. On analog silicon the intrinsic noise is not on the stored weight but in the *multiply-accumulate* — the forward pass, and hence the gradient, is noisy. We therefore test the mechanism in that realization: additive noise on the pre-activations (Gaussian by construction here — our tractable stand-in for the analog MAC’s additive trial-to-trial noise, whose additivity E5 measured but whose exact distribution we do not) (Fig. 8).

The mechanism survives, and it is the same signature. With forward noise, the barrier-conditioned rule is an inverted-U (lift 12.9 pts at $\sigma^* = 0.6$, paired Wilcoxon $p = 0.004$, 8 seeds), while the matched OU control is flat (0.0 pts; Fig. 8a). The noise the chip provides for free is the right kind of noise.

One stability fix is load-bearing. Porting to hardware first failed: the diagonal-Fisher importance is heavy-tailed, so the anchored drift blows the weights up and both methods collapse to chance. Normalizing and *clamping* the importance fixes it; without the clamp, retention is a flat 0.494 (chance) at every noise level (Fig. 8a, dotted), with the clamp it recovers to 0.797. We report this because it is the difference between the mechanism working on device and not.

The optimum is tunable to the device. The retention optimum’s noise level is set by the Doob-steering coupling: increasing it moves σ^* down to 0.05 (Fig. 8b), i.e. into the few-percent-CV band the measured BSS-2 noise occupies (§4.6). So the mechanism does not require a specific noise amplitude — it can be matched to a given chip’s intrinsic noise. This is what makes the on-chip realization reachable; we carry it out on real silicon in §4.8.

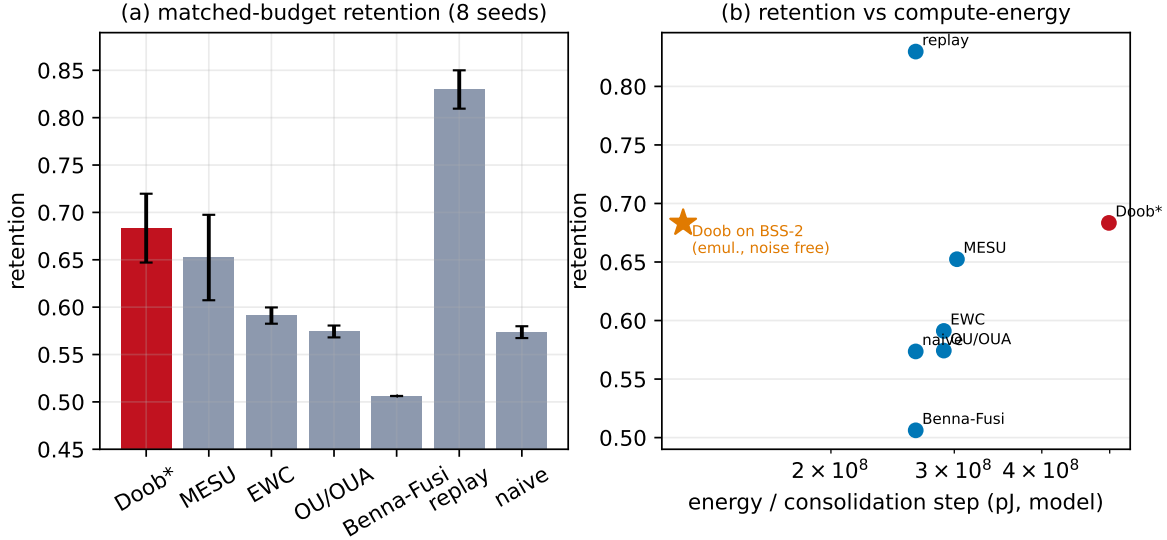


Figure 4: **Baselines (Split-MNIST, 8 seeds)**. (a) At its noise optimum the barrier-conditioned rule (Doob*, 68.3%) is the strongest *rehearsal-free* method: it matches MESU (65.2%) and significantly beats OU, EWC and naive (our Benna-Fusi did not learn in this configuration, 50.6% $\approx$ chance — flagged, not counted). Plain replay stores exemplars and scores higher (83.0%) — a different, memory-based budget class, shown for reference. (b) Retention vs. modelled compute-energy: a digital accelerator spends a modelled 23% (range 4%–41%) of each step *generating* the diffusion noise; on analog silicon that generation is free (the barrier steering still costs).

4.8 E7 — on-silicon demonstration (hardware-in-the-loop)

We ran the barrier-conditioned consolidation on real BrainScaleS-2 silicon, with the chip in the training loop: the analog-MAC forward pass executes on the device (hxtorch), so the intrinsic device noise perturbs the forward pass and hence the gradient — the diffusion is the chip’s own noise. A network learns a first continual-Yin-Yang task, then a second (a 90° rotation, strong interference) at the chip’s maximum intrinsic noise ($\text{avg}=1$), once with the barrier-conditioning and once without (the matched OU control), and we measure how much of the first task survives (Fig. 9).

Result. On silicon, the barrier-conditioning retains the prior task at 69.6% versus 54.0% for the unconditioned control — a 15.6-point retention gain from steering the chip’s own noise. This is the mechanism’s load-bearing claim, realized on the device a GPU cannot emulate for free: the intrinsic analog noise, conditioned on the memory barrier, consolidates a memory the control loses.

Honest scope. This is a single-seed proof of concept at one operating point (79 min of chip time), not a tuned optimum or a statistical study. The rule here leans toward stability: it pays for the retention with current-task plasticity (task 2 46.6% vs. 64.0%), so the two-task *average* accuracy is essentially matched (ours 58.1% vs. the control’s 59.0%) while retention strongly favours ours. This is a stability-plasticity shift, not a net-accuracy win, at an un-tuned point — one E6 (Fig. 8) shows can be balanced. We measured retention, not joules. A multi-seed noise sweep tracing the full on-silicon inverted-U, with measured energy, is the natural next study; the mechanism itself now demonstrably operates on the chip.

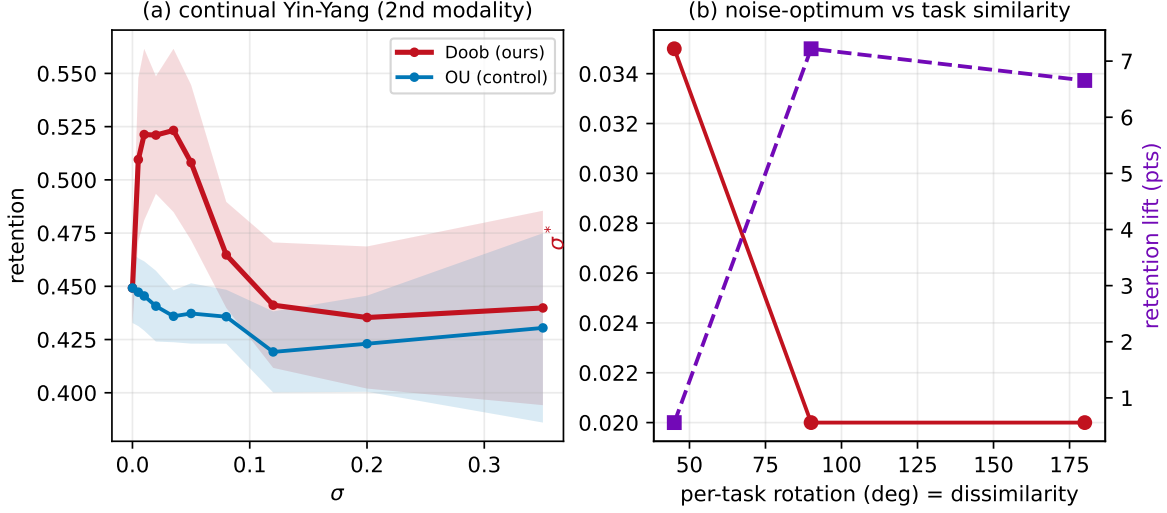


Figure 5: **Second modality.** (a) On continual Yin-Yang the inverted-U reproduces (lift 7.4 pts at $\sigma^* = 0.035$); the OU control is flat. (b) As tasks become less similar (larger per-task rotation, more interference) the noise optimum shifts, mapping the mechanism’s operating regime.

5 What we do not claim

- **The drift is not novel.** $-s(w - \mu)$ is a re-derived limit of OUA/MESU/EWC [Garcia Fernandez et al., 2024, Bonnet et al., 2025, Kirkpatrick et al., 2017]. Our contribution is the Doob barrier-conditioning (a) and the inverted-U (b), only in conjunction.
- **On-chip demonstration is a single-seed proof of concept.** We *did* train on real BrainScaleS-2 silicon with the chip in the loop (§4.8) and measured a retention gain, but at one seed and one operating point — not a tuned optimum, not a statistical sweep, and with the stability-plasticity trade stated. We measured retention, *not* joules (the energy numbers remain an operation-count model). A multi-seed on-silicon noise sweep with measured energy is the natural next study, not a claim we make here.
- **Benefit only at the optimum.** We do not claim the mechanism helps at arbitrary noise; it helps in an inverted-U window and hurts outside it. We report the whole curve, including the down-slope.
- **Not a retention SOTA claim.** The result is a mechanism and a signature, not a leaderboard entry. In particular plain replay, which stores raw exemplars, retains more than our rule (83.0% vs. 68.3%); we report this openly. Our comparison of record is against *rehearsal-free* anchored-drift methods under identical noise injection, where the signature and the standing are.

6 Limitations

Small single-head MLP testbeds; a diagonal-Fisher barrier; a ground-state (infinite horizon) h -transform rather than a finite-horizon survival function; a finite-force cap that removes the noise (σ^2) amplification of the steering where it binds (the tight-barrier tail — 8.8% of Doob steps at the operating point); a single-seed, single-operating-point on-silicon demonstration with energy

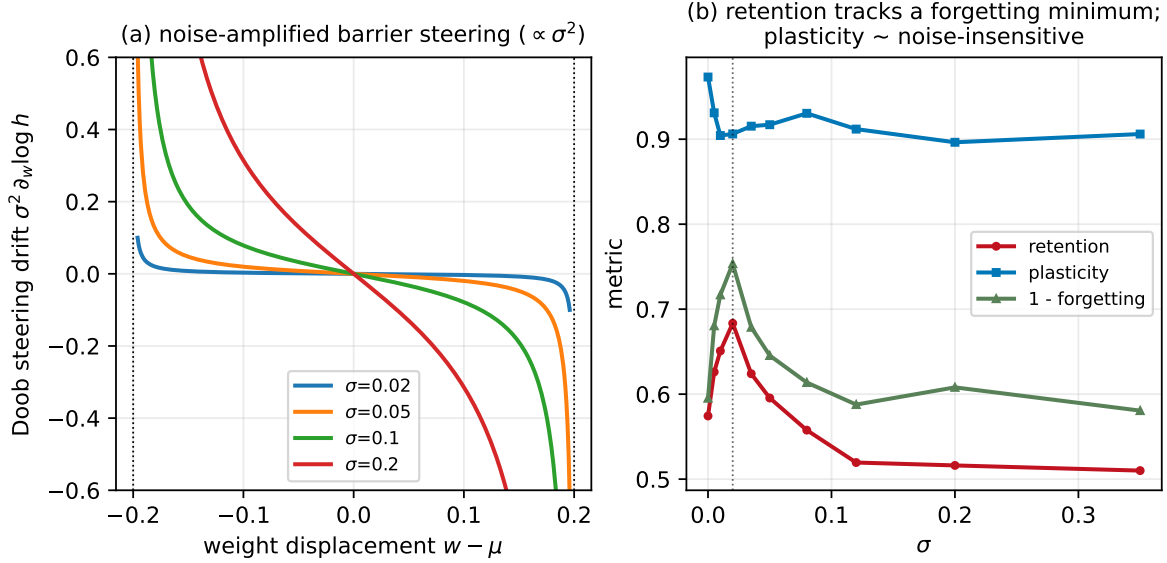


Figure 6: **Mechanism.** (a) The Doob steering drift $\sigma^2 \partial_w \log h$ for several noise levels: a barrier-divergent restoring force whose strength scales with the noise variance. (b) The retention optimum coincides with a forgetting minimum (1–forgetting peak) at σ^* ; plasticity is roughly noise-insensitive. The inverted-U is a protection effect, not a plasticity tradeoff.

modelled rather than measured; and no test of the device-noise distribution beyond additivity and a trial-to-trial independence check. Each is a place a larger study could overturn or extend the result; the pre-registered kill conditions name the failure modes.

7 Reproducibility

All code, `PLAN.md`, the committed `results/*.json`, and the figure/number generators are released. The pre-registration (GATE F, K1–K3, the fixed operating point, the noise grid, the inverted-U test) is git-verifiable for **E0–E4**: `PLAN.md` was committed before those results. **E5–E7** were added *after* BrainScaleS-2 access was obtained mid-project; they are post-access additions (dated in `PLAN.md`’s deviations log), not pre-registered in the same git-verifiable sense. Every number is a machine-generated macro (`numbers.tex` from `gen_paper_numbers.py`), and `verify_regen.py` checks it is byte-identical on regeneration. DOI at submission.

8 Conclusion

Casting per-synapse consolidation as a Doob h -transform makes a sharp, falsifiable prediction — intrinsic noise should non-monotonically improve retention — that the anchored-drift methods whose drift we share cannot make. The prediction holds in simulation, in a device-faithful emulation, and in the hardware-faithful forward-noise realization, is isolated to the barrier conditioning, and — on real BrainScaleS-2, with the chip’s own noise in the training loop — consolidates a memory the matched control loses. So intrinsic analog noise can be a consolidation resource that gets *cheaper* as devices get noisier, the opposite of the usual tax. What remains is to make that on-silicon demonstration a full study — a multi-seed noise sweep tracing the inverted-U on the device, with

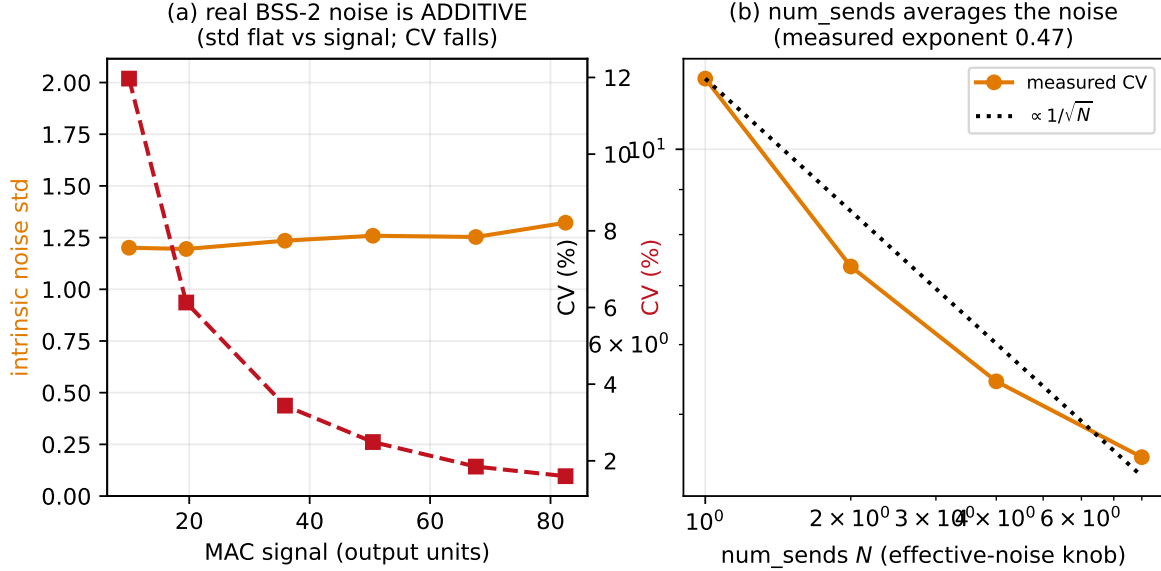


Figure 7: **Intrinsic noise measured on BrainScaleS-2 silicon** (chip hxcube7fpga3chip61.1). (a) The trial-to-trial noise std is flat in the signal (*additive*, 94.6%); the CV therefore falls as the signal grows. (b) `num_sends` averages the relative noise as $\approx 1/\sqrt{N}$ (measured exponent 0.47) — the on-chip effective-noise knob.

measured joules — and we have been careful to mark exactly where the single-seed proof of concept ends and that study begins.

References

- Marcus K. Benna and Stefano Fusi. Computational principles of synaptic memory consolidation. *Nature Neuroscience*, 19:1697–1706, 2016. doi: 10.1038/nn.4401.
- Djohan Bonnet, Kellian Cottart, Tifenn Hirtzlin, Tarcisius Januel, Thomas Dalgaty, Elisa Vianello, and Damien Querlioz. Bayesian continual learning and forgetting in neural networks. *Nature Communications*, 16, 2025. doi: 10.1038/s41467-025-64601-w. arXiv:2504.13569; introduces MESU (Metaplasticity from Synaptic Uncertainty), Eq. 11.
- Wei Deng, Yu Chen, Nicole Tianjiao Yi, Ye Yuan, et al. Reflected schrödinger bridge for constrained generative modeling. In *Uncertainty in Artificial Intelligence (UAI)*, 2024. arXiv:2401.03228.
- Yuanqi Du, Michael Plainer, Rob Brekelmans, Chenru Duan, Frank Noé, Carla P. Gomes, Alán Aspuru-Guzik, and Kirill Neklyudov. Doob’s lagrangian: A sample-efficient variational approach to transition path sampling. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.07974.
- Jesus Garcia Fernandez, Nasir Ahmad, and Marcel van Gerven. Ornstein-uhlenbeck adaptation as a mechanism for learning in brains and machines. *Entropy*, 26(12):1125, 2024. doi: 10.3390/e26121125. arXiv:2410.13563.
- Jeremy Heng, Valentin De Bortoli, Arnaud Doucet, and James Thornton. Simulating diffusion bridges with score matching. *arXiv preprint arXiv:2111.07243*, 2021.
- Mark Horowitz. 1.1 computing’s energy problem (and what we can do about it). *IEEE International Solid-State Circuits Conference (ISSCC)*, pages 10–14, 2014. doi: 10.1109/ISSCC.2014.6757323.

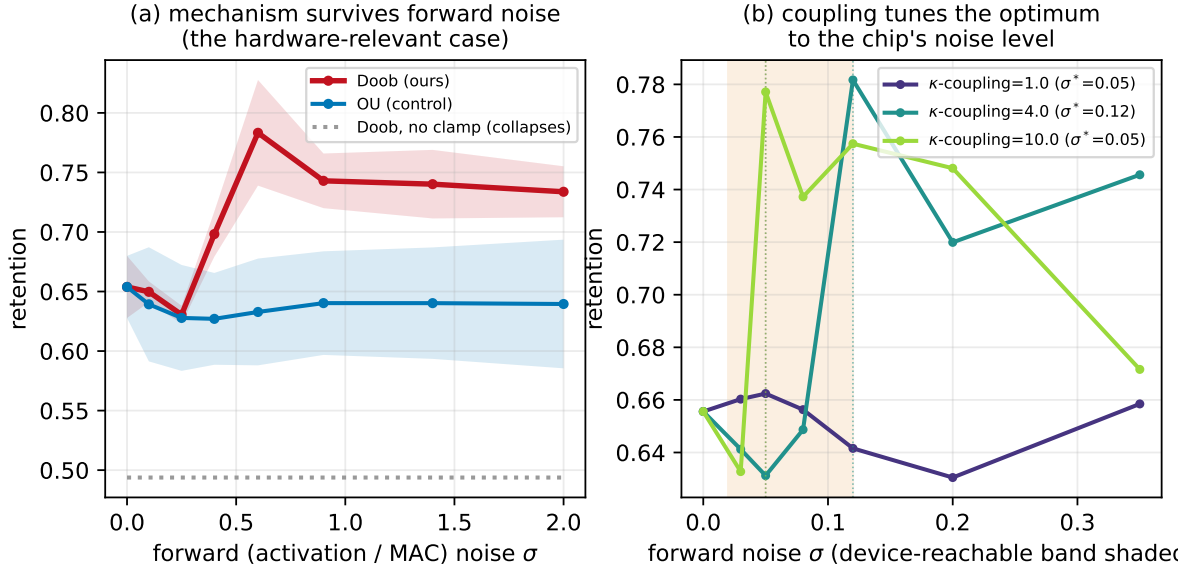


Figure 8: **Forward-noise (hardware-faithful) realization.** (a) With noise in the forward pass, Doob is an inverted-U (lift 12.9 pts, $p = 0.004$) and OU is flat; *without* the importance clamp both collapse to chance (dotted). (b) The Doob-steering coupling tunes the retention optimum down to $\sigma^* = 0.05$ — into the device-reachable noise band (shaded) — so the mechanism is portable to a given chip’s intrinsic-noise level.

James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017. doi: 10.1073/pnas.1611835114.

Ivan Kolesnikov and Nadezhda Semenova. Internal noise in hardware deep and recurrent neural networks helps with learning. *arXiv preprint arXiv:2504.13778*, 2025.

Laura Kriener, Julian Göltz, and Mihai A. Petrovici. The yin-yang dataset. *Neuro-Inspired Computational Elements (NICE)*, 2022. arXiv:2102.08211.

Luca Manneschi, Ian T. Vidamour, Kilian D. Stenning, et al. Noise-aware training of neuromorphic dynamic device networks. *Nature Communications*, 16, 2025. doi: 10.1038/s41467-025-64232-1. arXiv:2401.07387.

Toan Nguyen, Kien Do, Duc Kieu, and Thin Nguyen. h-edit: Effective and flexible diffusion-based editing via doob’s h-transform. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. arXiv:2503.02187.

Christian Pehle, Sebastian Billaudelle, Benjamin Cramer, Jakob Kaiser, Korbinian Schreiber, Yannik Stradmann, Johannes Weis, Aron Leibfried, Eric Müller, and Johannes Schemmel. The brainscales-2 accelerated neuromorphic system with hybrid plasticity. *Frontiers in Neuroscience*, 16:795876, 2022. doi: 10.3389/fnins.2022.795876.

Nimrod Shaham, Jay Chandra, Gabriel Kreiman, and Haim Sompolsky. Stochastic consolidation of lifelong memory. *Scientific Reports*, 12:13107, 2022. doi: 10.1038/s41598-022-16407-9.

Johannes Weis, Philipp Spilger, Sebastian Billaudelle, Yannik Stradmann, Arne Emmel, Eric Müller, Oliver Breitwieser, Andreas Grübl, Joscha Ilmberger, Vitali Karasenko, Mitja Kleider, Korbinian Schreiber, Christian Weilbach, and Johannes Schemmel. Inference with artificial neural networks on analog neuromorphic hardware. In *IoT Streams for Data-Driven Predictive Maintenance / ITEM (Springer CCIS 1325)*, 2020. arXiv:2006.13177.

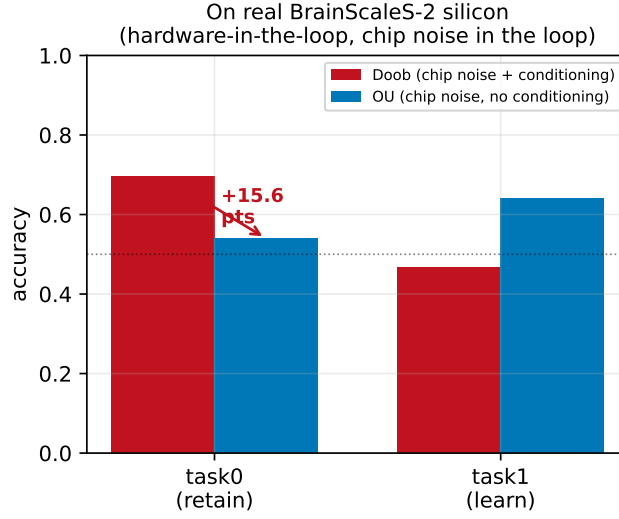


Figure 9: **On real BrainScaleS-2 silicon** (hardware-in-the-loop; the chip’s intrinsic noise is the diffusion). After learning task 2, the barrier-conditioned rule retains task 1 at 69.6% vs. 54.0% for the matched unconditioned control (+15.6 pts). Single seed, one operating point; the rule trades some task-2 plasticity for the retention (stated in text).

Fatima Tuz Zohora, Vedant Karia, Nicholas Soures, and Dhireesha Kudithipudi. Probabilistic metaplasticity for continual learning with memristors. *Scientific Reports*, 14, 2024. doi: 10.1038/s41598-024-78290-w. arXiv:2403.08718.